Characterization of a Brent–Kung Tree Adder: A Study on Performance and Efficiency Metrics

Amit Barman (SR No: 04-02-01-10-51-25-1-27002)
Department of Electrical Communication Engineering, IISc Bangalore, India
amitbarman@iisc.ac.in

A. Assignment Objectives

The objective of this assignment is to design and analyze a Brent–Kung Tree adder architecture. The study focuses on evaluating its critical path delay, area, and power characteristics to assess overall performance and efficiency. The results will help in understanding the trade-offs between speed, hardware complexity, and power consumption in high-performance adder designs.

I. Schematic of Brent-Kung Tree Adder

The Brent–Kung adder employs a logarithmic carry computation structure, achieving a delay of $O(log_2 N)$ for an N-bit addition. The prefix network is designed with balanced stages that progressively combine generate (G) and propagate (P) signals to compute carries efficiently. Compared to other parallel prefix adders, such as Kogge–Stone or Sklansky, the Brent-Kung design uses fewer wiring tracks and logic nodes, reducing area and power consumption while maintaining reasonable speed.

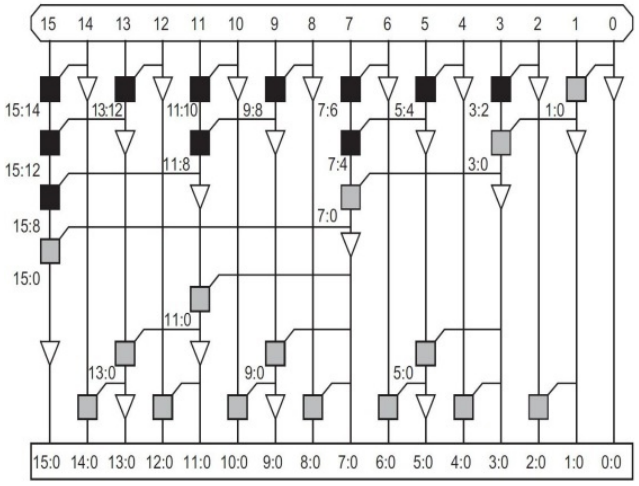


Fig. 1. Block diagram design of Brent-kung Tree Adder

The schematic demonstrates $2log_2 N - 1$ logic levels, with a trade-off between reduced fan-out and moderate delay. This makes the architecture well-suited for VLSI implementations, where interconnect complexity and power efficiency are critical considerations. The design's hierarchical structure also simplifies routing and enhances scalability for higher bit-width adders.

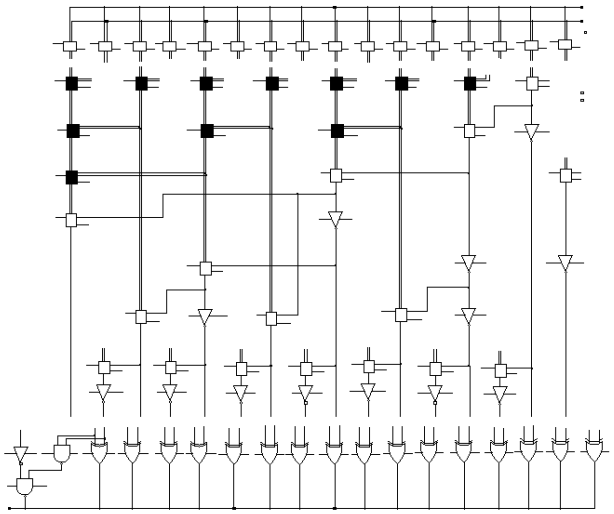


Fig. 2. Schematic of Brent-Kung Tree Adder

II. Layout of Brent-Kung Tree Adder

The layout of the Brent–Kung adder is designed to optimize area utilization and signal routing. Standard cells are arranged to maintain short interconnect paths between prefix stages, minimizing delay and parasitic effects. Symmetrical placement of generate and propagate logic ensures balanced signal flow. A total of 5 types of metal lines, i.e Metal1, Metal2, Metal3, Metal4 and Metal5 are used for interconnect. The final layout achieves a compact structure with reduced wire congestion and optimized performance for area and power efficiency

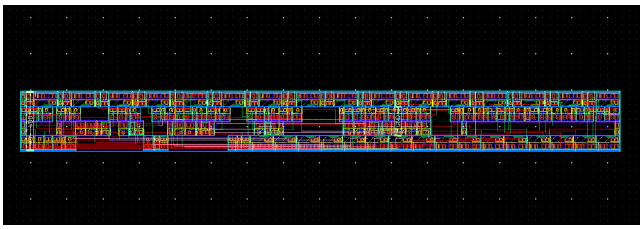


Fig. 3. Layout view of Brent-Kung Tree Adder

III. Characterization of 16-bit Brent-Kung Tree Adder

The Brent–Kung adder was characterized in terms of delay, power, and energy under different process corners — Typical-Typical (TT), Slow-Slow (SS), and Fast-Fast (FF). A $2fF$ capacitor and a minimum delay inverter load were used to model the output capacitance for accurate delay estimation. The critical path delay was measured by applying input transitions and observing the propagation of carry signals through the prefix network.

The Total Area of our layout of the Brent-Kung Tree adder is **695.552** μm^2. The worst-case critical path delays are $549.487ps$ (Before PEX) and $1.1903ns$ (After PEX).

TABLE I
Characterization of Brent-Kung in terms of Delay, Power and Energy at various corners

Corner	Temp ($^\circ C$)	Supply Volt (V)	Delay(t_{pd}), after PEX	Power consumption (μW)	Energy (pJ)
FF	0	1.1	805.506ps	40.704	4.074
SS	100	0.9	2.723ns	26.321	2.632
TT	27	1	1.1903ns	32.850	3.285

The Delays and powers are calculated by considering the load capacitance of $C = 2fF$ and a unit inverter at the output.

A. Hand Calculation of delays

From the schematic, we have got that in critical path there are a PG generator block, 6 Grey cell, 2 NOT gate, and a XOR gate. So, Delay is ,

$$t_d = t_{PG} + 6t_{GrayCell} + 2t_{NOT} + t_{XOR} \tag{1}$$

As we know, $t_{PG} = t_{XOR}$ so,

$$t_d = 6t_{GrayCell} + 2t_{NOT} + 2t_{XOR} \tag{2}$$

$$t_{GrayCell} = t_{NAND} + t_{NOR} + t_{NOT} \tag{3}$$

$$t_d = 6 \times 51.173 + 2 \times 13.155 + 2 \times 59.301 \tag{4}$$

$t_d = 451.95ps$

This value is close to the worst-case critical path delay of $549.48ps$.

References

[1] Donald A. Neamen, Semiconductor Physics and Devices: Basic principles – 4th Ed.

[2] Weste, N.H.E. and Harris, D.M., CMOS VLSI Design: A Circuits and Systems Perspective, (2011), Addison Wesley, books.google.co.in/books?id=sv8OQgAACAAJ

[3] Class Notes of DVLSI